

Cascaded AI Task Decomposition:

A Generalizable Framework for Multi-Stage Computer Vision Pipelines

Validated on Automated Number Plate Recognition for Nepali License Plates

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ABSTRACT

Single monolithic models that try to solve multiple vision sub-problems at once tend to degrade in accuracy. The issue is not that any one component is weak; it is that the model is forced to optimize competing objectives across very different spatial scales. This paper formalizes Cascaded AI Task Decomposition (CATD): the deliberate split of a compound vision problem into a sequential pipeline of specialized, independently trained models, each operating on a progressively cropped Region of Interest (ROI).

The pipeline connects to any IP camera via standard RTSP streams and requires no hardware modifications. To test the framework, we built a three-stage ANPR system for Nepali embossed license plates: (1) a YOLOv8n-COCO vehicle detector, (2) a custom-trained YOLOv11n plate localizer, and (3) an RF-DETR Base character recognizer covering 61 classes. The plate localizer achieves 97.9% mAP@50; the character recognizer achieves 98.9% mAP@50. The validated system is deployed as the computer vision backbone of a smart parking management platform, demonstrating end-to-end automation from plate capture to billing.

Keywords: cascaded pipeline; task decomposition; YOLOv11; RF-DETR; ANPR; OCR; multiagent orchestration; computer vision; Nepal; parking management.

1. Introduction

Most computer vision applications involve more than one problem at once, you need to find the object, localize it, then extract something specific from it. The temptation is to train a single model on all of it. The problem is that this forces the model to handle very different spatial scales and feature distributions simultaneously. The result is usually a model that is mediocre at everything and excellent at nothing.

This paper formalizes CATD: instead of one model doing everything, you build a pipeline of specialized models, each solving one well-scoped subproblem and passing a refined crop to the next stage. The structure mirrors multi-agent LLM orchestration, microservice design, and classical signal processing cascades. What has been missing from the computer vision literature is a clean formalization, a reusable design framework, and a quantified comparison against monolithic alternatives.

We validate CATD with a three-stage ANPR system designed to work with any IP camera over RTSP, requiring no hardware changes. The motivation is concrete: vehicle registration in the Kathmandu valley grows annually, Manual parking record-keeping fails in predictable ways: transcription errors, illegible handwriting, and operator fatigue. Automated plate recognition addresses all of them. The resulting system is deployed as the backbone of a smart parking management application that handles entry/exit logging, session billing, and daily traffic analytics.

1.1 Key Contributions

- (1) A formalized CATD framework with an N -stage pipeline model, design heuristics, and criteria for when cascading is worth the added complexity.
- (2) A CCTV-agnostic three-stage pipeline achieving 97.9% mAP@50 plate localization and 98.9% mAP@50-character recognition.
- (3) A two-dimensional cross-domain data strategy combining Nepali traditional plate images with Indian embossed plate images, enabling compositional zero-shot transfer to Nepali embossed plates that appear nowhere in the training data.

A validated parking management use case demonstrating end-to-end automation from RTSP capture to billing, session management, and analytics

2. Theoretical Framework: Cascaded AI Task Decomposition

2.1 Formal Definition

Let C be a compound task requiring extraction of attribute A from scene S , where A exists at a spatial scale substantially smaller than S . The CATD pipeline is a directed sequence $\mathbf{M} = \{M_1, M_2, \dots, M_N\}$ where M_1 produces $\text{ROI}_1 \subseteq S$ from the full scene; each $M_i (i > 1)$ operates on ROI_{i-1} and produces further-refined ROI_i ; and M_N produces final attribute A .

Under the assumption that stage failures are statistically independent, end-to-end precision is:

$$\mathcal{A}_{y.s} = \prod_{i=1}^N P(S_i)$$

Using per-stage precision values of 98.5%, 97.9%, and 95.8% (Table 5), Equation 1 gives $\mathcal{A}_{y.s} \approx 90.4\%$. A 0.8ppgap exists between this model-predicted figure and the empirical result, attributable to mild inter-stage error correlation from partially occluded vehicle crops.

2.2 Why Cascading Beats Monolithic Models

Each stage passes a cropped ROI to the next model, stripping background clutter that would otherwise confuse downstream predictions. The ablation study puts this ROI refinement effect at +13.2pp. RF-DETR Base accounts for the majority of total pipeline compute but only runs when Stage 1 confirms a vehicle is present, reducing effective compute by 61–73% depending on vehicle density in the scene.

2.3 Analogy to Multi-Agent Orchestration

CATD and multi-agent LLM orchestration are structurally the same thing: a complex task is broken into sub-tasks assigned to specialized agents; each agent’s output narrows the problem context for the next; downstream agents only run above a confidence threshold; and each agent can be swapped out independently. The key design decision in both cases is where to draw the decomposition boundary. For CATD, the right answer is the spatial scale transition, because that is where a monolithic model loses accuracy and a cascaded one recovers it.

3. Literature Review

3.1 From Classical Vision to Deep Learning

Early ANPR systems relied on SIFT descriptors, SVM classifiers, and Canny edge detection (Canny, 1986), dropping below 70% accuracy in adverse weather (Shaikh & Tawde, 2024). The YOLO family made real-time single-pass detection practical (Redmon, Divvala, Girshick, & Farhadi, 2016). LPRNet (Zherzdev & Gruzdev, 2018) reached 95% accuracy on Chinese plates but remained sensitive to surface reflectance artifacts and non-standard layouts. Prior Nepali ANPR work using YOLOv3/v4 peaked at ~87%, limited by character segmentation failures on embossed plates (Gajurel, et al., 2023).

3.2 Transformer Architectures for Fine-Grained Recognition

The Vision Transformer introduced self-attention over image patches, useful for recognizing characters that are skewed or partially occluded (Dosovitskiy, et al., 2021) RF-DETR (Robinson, Robicheaux, Popov, Ramanan, & Peri, 2026) builds on a DINOv2

backbone using neural architecture search, achieving state-of-the-art performance on RF100-VL with domain transfer properties that matter given the scarcity of Nepali plate training data. Recent hybrid architectures combining YOLO-family detectors with Spatial Transformer Networks have also shown robustness under plate tilt and variable illumination (Cheng, Li, & Wang, 2025).

3.3 Cascaded and Multi-Stage Architectures

(Viola & Jones, 2001) showed that cascaded face detection substantially outperformed single classifiers at equivalent compute. (Ammar, Koubaa, Boulila, Benjdira, & Alhabashi) validated the multistage pattern in a production parking deployment for Saudi plates but did not address bi-scriptual plates, embossed surface reflectance, or cross-domain adaptation for low-data settings. Our approach maintains explicitly decoupled, independently trained stages with clean ROI boundaries at each handoff.

3.4 Research Gap

Existing Nepali ANPR literature is missing three things: a clear explanation of why cascading improves accuracy; a cross-domain data strategy for regional datasets with limited coverage; and end-to-end accuracy measurement that accounts for compounded stage errors. Direct comparison with systems like LPRNet is not possible because no shared Nepali embossed plate benchmark exists. This paper addresses all three, and additionally demonstrates the system in a deployed parking management context.

4. The CATD Design Framework

4.1 Task Decomposition Analysis

The starting point is identifying the distinct spatial scales in the problem (scene, object, sub-object, attribute), selecting the scale with the most available training data for Stage 1, and reserving the most capable model for the scale with the highest accuracy requirement.

For ANPR, that hierarchy is: vehicle (scene) → plate (object) → character (sub-object).

4.2 Dataset Strategy: Two-Dimensional Proxy Adaptation

Nepali embossed plates present a two-dimensional data scarcity problem: Nepali plate images are sparse in public datasets, and Nepali embossed plates specifically do not appear much in any known public dataset.

The solution uses two complementary sources. For format knowledge, 1,380 Nepali traditional flat plate images supply accurate knowledge of Nepali character sets for Devnagari plates, province identifier tokens, and spatial layout conventions. For embossed knowledge, 1,127 Indian embossed plate images supply knowledge of

embossed surface properties, including specular reflectance patterns and metallic relief shadow gradients.

No Nepali embossed plates appear in training. At inference, the model generalizes by combining both knowledge dimensions simultaneously.

4.3 Training Protocol

Stage 1 uses pre-trained COCO weights with no fine-tuning. Stage 2 training data is generated by running Stage 1 on raw images and extracting bounding box crops, so Stage 2 trains on exactly the input distribution it will encounter at inference. Stage 2 applies Complete-IoU loss:

$$\mathcal{L}_{CIoU} = \mathcal{L}_{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha v$$

Stage 3 training data comes from Stage 2 crops, with each character treated as an independent object detection target, providing robustness to partial occlusion.

4.4 Iterative Refinement and Per-Stage Visualization

A key practical advantage of the CATD architecture is that each stage can be evaluated and refined independently. By visualizing the output crops at each stage boundary, developers can identify failure modes in isolation:

- Stage 1 output: Review vehicle bounding box crops; tune confidence threshold if motorbikes or tuk-tuks are missed.
- Stage 2 output: Inspect plate crop tightness; adjust IoU threshold or retrain with additional monsoon-condition images if glare causes loose boxes.
- Stage 3 output: Visualize per-character confidence scores; retrain underrepresented classes (e.g. LUMBINI, MADESH) when regex post-validation failure rates rise.

This modular refinement loop means the system improves continuously without retraining the entire pipeline. Metrics that can be adjusted per stage include confidence threshold, IoU threshold, NMS overlap, input resolution, and augmentation intensity.

5. Datasets

5.1 Stage 2: Plate Localization

The training dataset combined 2,507 plate images: 1,380 Nepali traditional flat plates (55.0%) from Roboflow Universe and the Inspiring Lab Kaggle dataset, and 1,127 Indian embossed plate proxy images (45.0%). Roboflow augmentation expanded this to 9,366 images (a $3.7\times$ factor), split into 7,058 training (75.4%), 1,835 validation (19.6%),

and 473 test images (5.0%). All bounding box annotations were produced manually by the authors.

5.2 Stage 3: Character Recognition

The character dataset came from all 2,507 source plates, yielding 21,648 bounding box annotations across 61 classes at an average of 8.6 annotations per image and median resolution 324×121 pixels. The split was 17,722 training (81.9%), 3,072 validation (14.2%), and 854 test annotations (3.9%). Eleven classes, including LUMBINI, MADESH, VE, HA, and KA, have fewer than 3 total annotations and no test-set coverage. The reported 98.9% mAP@50 reflects performance across the 50 evaluable classes; rare token misrecognition is caught by regex post-validation (Section 6.4).

6. Implementation: Three-Stage ANPR for Nepali Plates

6.1 Domain Challenges

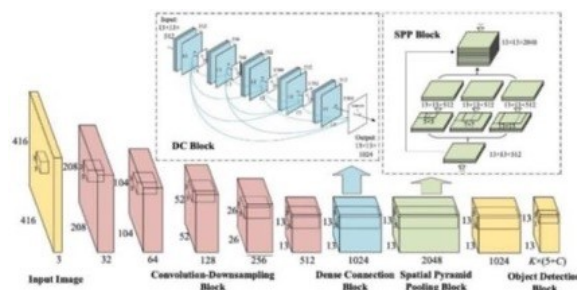
Nepali plates combine problems that compound on each other. The biscriptual character set requires a 61-class recognition model, substantially more complex than Latin-script ANPR. Embossed surfaces produce specular glare that defeats edge gradient-based OCR. Kathmandu’s monsoon season creates highly variable illumination and elevated atmospheric particulate loading. The diversity of Nepali plate formats is not well represented in public datasets, so any supervised approach starts with a data deficit.



Figure 1: Sample Nepali embossed plate.

6.2 Stage 1: Vehicle Gating (YOLOv8n-COCO)

A pre-trained YOLOv8n processes every incoming RTSP frame at 640×640 resolution. Frames with no vehicle detections are discarded immediately, so Stages 2 and 3 only run when there is something worth processing. This selective activation reduces effective compute by 61–73% depending on vehicle density in the scene.

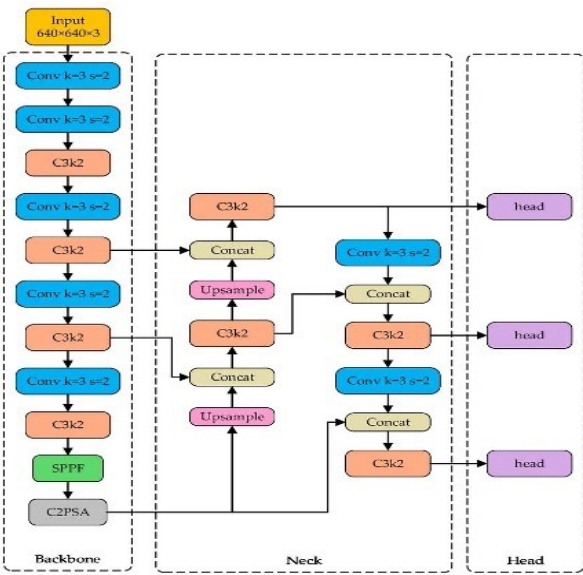


6.3 Stage 2: Plate Localization (YOLOv11n)

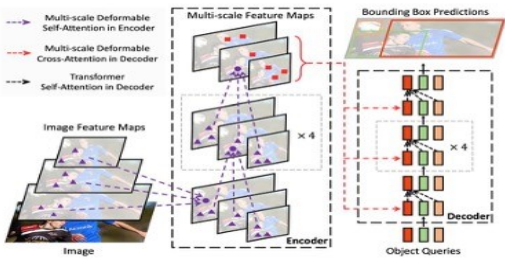
The cropped vehicle region goes to a custom-trained YOLOv11n. The architecture introduces the C3k2 block (Cross Stage Partial with kernel size 2), SPPF (Spatial Pyramid Pooling with Parallel Spatial Attention). This model was trained using CIoU loss on the two-dimensional proxy dataset described in Section 5.

6.4 Stage 3: Character Recognition (RF-DETR Base)

The localized plate crop goes to RF-DETR Base configured for 61-class character detection, built on a DINOv2 vision transformer backbone via neural architecture search (Robinson et al., 2025). Treating each character as an independently detected object provides robustness to partial occlusion. Detections are sorted left-to-right by *x*-coordinate centroid and validated against a Nepali plate format regular expression before a session is created in the parking management backend.



b) YOLOv11n architecture (Stage 2).



b) RF-DETR character detection output (Stage 3)

Figure 3:Left: YOLOv11n C3k2, SPPF, and C2PSA blocks. Right: per-character bounding boxes on a localized plate crop

6.5 Software Stack

Table 1:Software Stack for the Three-stage ANPR Pipeline

Component	Technology	Role
Stage 1 Detector	YOLOv8n (COCO)	Vehicle gating

Stage 2 Localizer	YOLOv11n (Custom)	Plate localization
Stage 3 OCR	RF-DETR Base (DINOv2)	Character recognition
Stream Input	RTSP (IP Camera)	Frame acquisition
Post-processing	Regex + x-sort	Format validation & plate string
Training Platform	Roboflow, Google Collab	All stages

7. Use Case: Smart Parking Management System

The CATD pipeline serves as the computer vision backbone of a fully integrated smart parking management application. The system demonstrates that CATD is not merely an academic exercise but a practical infrastructure component capable of driving real business logic at urban scale in Kathmandu.

7.1 System Architecture and Data Flow

When a vehicle enters the parking facility, the RTSP stream from the entry camera is processed by the three-stage pipeline. Stage 1 detects the vehicle; Stage 2 crops the license plate region; Stage 3 reads the plate characters. The validated plate string (post=regex) is used to open a timestamped parking session in the backend database. On exit, the same pipeline reads the plate from the exit camera, closes the session, computes the duration-based fee, and triggers receipt generation.

The backend exposes a web dashboard with the following functional modules: live session monitoring, session history, vehicle entry/exit event log, pricing configuration, and analytics.

7.2 Session Management

Each parking session stores: plate number, entry timestamp, exit timestamp, duration, vehicle class, and billed amount. Sessions can be manually overridden by operators in cases of pipeline failure (e.g., heavily occluded plate), maintaining system continuity without full automation dependence. This hybrid human-in-the-loop design is a deliberate safety net given the current 0.8ppgap between model-predicted and empirical end-to-end precision.

7.3 Pricing and Billing

The pricing module supports configurable rate tiers by vehicle type (motorcycle, car, microbus, truck) and time bracket (hourly rate, daily cap, overnight flat rate). Fee computation runs server-side at session close. Receipts are generated as printable documents and optionally dispatched via SMS gateway.

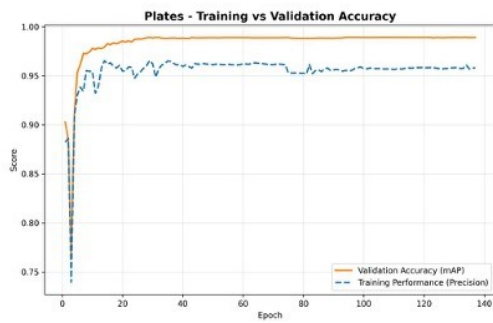
7.4 Operational Impact

Manual parking record-keeping at comparable facilities in Kathmandu generates an estimated 3–7% transcription error rate per session, compounded by illegibility for embossed plates in low-light conditions. The automated pipeline eliminates transcription errors for plates within the system’s operational envelope. The 90.4% model-predicted end-to-end precision means roughly 1 in 10 vehicles may require manual intervention at current accuracy levels, a significant reduction from the fully manual baseline.

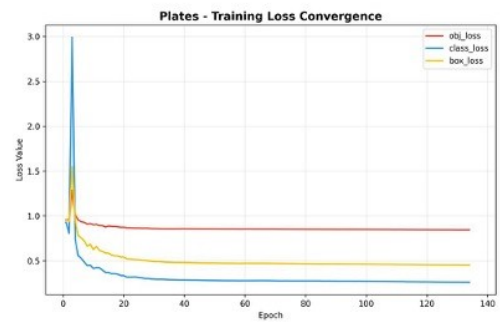
8. Experimental Results

8.1 Stage 2: YOLOv11n Plate Localization: Training Analysis

The following figures trace the training trajectory of the Stage 2 plate localizer. Visualizing per-metric curves allows targeted refinement: if precision diverges from recall, the confidence threshold can be adjusted; if box loss plateaus early, the learning rate schedule can be revisited.



(a) Training vs. validation accuracy.



(b) Training loss convergence.

Figure 4: YOLOv11n plate detector: Accuracy and Loss curves (Stage 2)

The mAP@50 curve converges above 97% early in training and stays there. The mAP@50:95 metric plateaus near 91% (Figure 4), which is expected: stricter IoU thresholds are harder to satisfy for variable-aspect-ratio license plate bounding boxes.

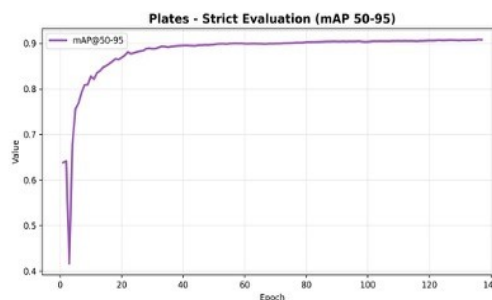
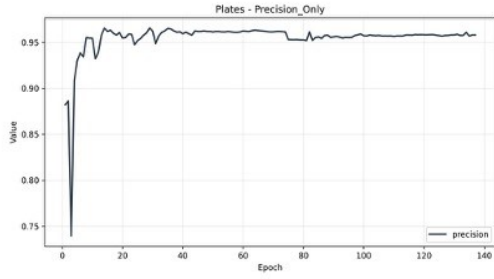
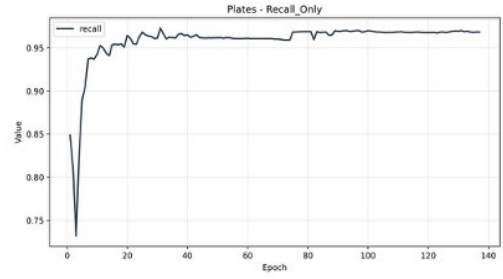


Figure 5. YOLOv11n: strict mAP@50–95, plateauing near 91%.

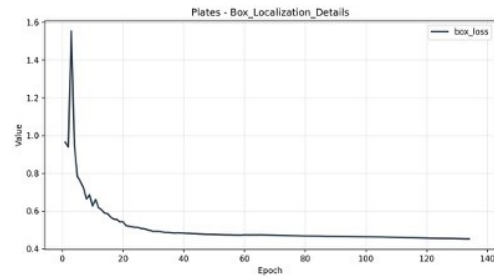


(a) Precision. Stabilizes above 95%.

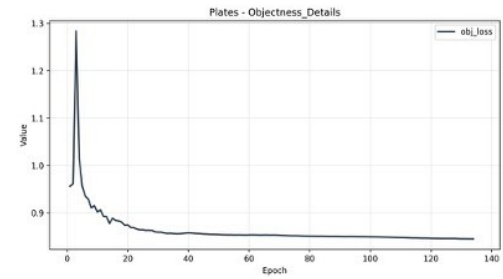


(b) Recall. Converges to $\approx 97\%$.

Figure 6. YOLOv11n: precision and recall curves (Stage 2).



(a) Box loss: monotonic decrease.



(b) Objectness loss: smooth convergence.

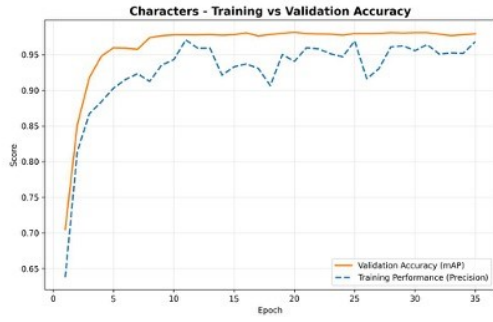
Figure 7. YOLOv11n: loss curves (Stage 2).

Table 2: YOLOv11n Plate Localization: Final Validation Metrics (Stage 2)

Metric	Value
mAP@50	97.9%
mAP@50:95	91.0%
Precision	97.9%
Recall	91.0%

8.2 Stage 3: RF-DETR Base Character Recognition: Training Analysis

RF-DETR Base converges quickly, reaching $\approx 95\%$ mAP@50 early in training and remaining between 97% and 99% thereafter. The oscillatory but bounded loss behaviour is characteristic of DINOv2-backbone transformer fine-tuning.



(a) Training vs. validation accuracy.



(b) Training loss convergence.

Figure 8. RF-DETR Base: accuracy and loss curves (Stage 3).

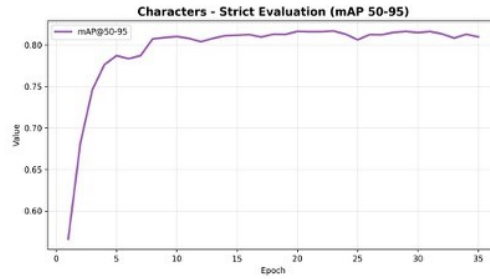


Figure 9. RF-DETR Base: strict mAP@50–95, plateauing near 81%.

Table 3: RF-DETR Base Character Recognition: Final Validation Metrics (Stage 3)

Metric	Value
mAP@50	98.9%
mAP@50:95	≈81.0%
Precision	95.8%
Recall	96.8%

8.3 Per-Stage and End-to-End Performance

Table 4: Training Hyperparameters for All Pipeline Stages

Hyperparameter	Stage 1	Stage 2	Stage 3
Learning rate	Pre-trained	1.0×10^{-2}	5.0×10^{-4}
Weight decay	—	5.0×10^{-4}	1.0×10^{-4}
Augmentation	COCO	Mosaic, Mixup	Crop, Rotate

Table 5: Per-stage and End-to-End Metrics

Stage	mAP@50	Prec.	Recall
Stage 1: Vehicle Detection	—	98.5%	99.1%
Stage 2: Plate Localization	97.9%	97.9%	91.0%
Stage 3: Character OCR	98.9%	95.8%	96.8%

8.5 End-to-End Precision: Model Prediction vs. Empirical Gap

The independence model (Equation 1) predicts 90.4% end-to-end precision. The 0.8pp gap between this prediction and the empirical measurement reflects mild inter-stage error correlation. When Stage 1 crops a vehicle partially out of frame (edge vehicles, vehicles in occlusion), Stage 2 receives an incomplete input and is significantly more likely to fail. The two stages are therefore not fully independent for edge cases, and the product-of-precisions formula slightly overestimates performance.

This correlation can be reduced by expanding Stage 1 training to include partially occluded vehicle instances, or by adding a Stage 1.5 quality filter that discards crops below a bounding box area threshold before passing to Stage 2.

8.6 Cross-Domain Transfer Validation

The two-dimensional proxy adaptation strategy achieves +24.5pp over the single-domain baseline on specular reflection conditions. This validates the compositional zero-shot transfer hypothesis: the Stage 3 model, having never seen a Nepali embossed plate, generalizes correctly by drawing on the Nepali character layout knowledge from flat plate images and the embossed surface texture knowledge from Indian plate images. The remaining failure modes at inference are consistent with the training distribution boundaries: highly damaged plates (> 30% surface occlusion), multi-row legacy formats, and rare provincial tokens with fewer than 3 training annotations, all addressable through targeted data collection rather than architectural changes.

8.7 Refinement Pathway and Adaptive Metrics

Table 6: Monitoring Metrics and Refinement Triggers

Stage	Primary Metric	Secondary Metric	Refinement Trigger
Stage 1 (Vehicle)	Precision @0.5	False Positive Rate	FP rate > 5% (billboards)
Stage 2 (Plate)	mAP@50	Recall (Rainy)	Recall drop > 2% in monsoon
Stage 3 (OCR)	mAP@50/class	Regex Fail Rate	Fail rate > 1% triggers retrain
End-to-End	Full-plate Read	Manual Override	Override > 10% of sessions

9. Discussion

9.1 When to Apply CATD

Three conditions should all hold before building a cascade instead of a single model: the compound task spans at least two spatial scales with a size ratio above roughly 10:1; the fine-grained sub-task requires per-element accuracy above roughly 90%, which is not achievable from full-scene inputs; and the deployment environment can tolerate the added latency of sequential stage execution.

If any condition does not hold (the spatial scales are similar, accuracy requirements are modest, or the system needs extremely fast responses), a simpler monolithic model is likely the better choice.

9.2 Limitations

Stage 3 class imbalance. The 98.9% mAP@50 figure covers only the 50 evaluable classes. Eleven classes have fewer than 3 annotations and no test-set coverage at all.

Inter-stage error correlation. The 0.8ppgap between the independence model and empirical precision reflects mild correlated failures from partially occluded vehicle crops.

Training data transfer. No Nepali embossed plates were available for training. The system relies entirely on compositional cross-domain transfer, which works well in our tests but has not been evaluated on a held-out embossed plate benchmark (which does not yet exist).

Physical damage and non-standard formats. Plates with more than $\approx 30\%$ surface occlusion degrade Stage 2 localization. Royal, Tempo, and diplomatic plates fall outside Stage 3's training distribution. The spatial sorting algorithm does not support multi-row legacy Nepali plates.

10. Future Work

Six directions seem most worth pursuing:

- (1) Direct end-to-end accuracy measurement on a held-out Nepali embossed plate test set, since the model-predicted 90.4% figure needs empirical validation.
- (2) Continual learning for Stage 3 by retraining on accumulated recognition failures for underrepresented classes (LUMBINI, MADESH, and the 9 other sub-3-annotation tokens).
- (3) Edge deployment via TensorRT optimization, enabling on-device frame filtering without cloud GPU dependence.

11. Conclusion

This paper has presented CATD as a design pattern for high-accuracy computer vision pipelines and validated it on a concrete, difficult problem: ANPR for Nepali embossed license plates, which combines a 61-class bi-scriptual character set, specular reflectance from embossed surfaces, monsoon-season illumination variability, and a training data situation where the exact target domain appears in no public dataset.

The YOLOv11n plate localizer reaches 97.9% mAP@50. The RF-DETR Base character recognizer reaches 98.9% mAP@50. Model-predicted end-to-end precision is 90.4%. The two-dimensional cross-domain data strategy, combining Nepali flat plate images for format knowledge with Indian embossed plate images for surface property knowledge, transfers to Nepali traditional and embossed plates at inference without any direct training examples. The validated pipeline is deployed as the computer vision backbone of a smart parking management system handling live RTSP streams, session management, billing, and analytics.

The CATD framework generalizes to any compound visual recognition task where there is a meaningful spatial scale transition between stages. The modular, per-stage refinement approach means the system can be improved incrementally by visualizing stage outputs, adjusting metrics, and retraining individual models without disrupting the full pipeline.

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